

Recognizing Operator Task States from Multimodal Physiological Signals in Industrial Human–Robot Collaboration: a Subject-Disjoint Evaluation

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Abstract. Robots that assist human operators in industrial settings must adapt to what the operator is doing. One source of evidence for the operator’s current state is their own physiology: brain and peripheral signals change systematically across cognitive, affective, and physical workload. In this paper we study whether the *type* of task an operator is performing can be recognized from such signals in a way that generalizes to operators never seen during training. We work with the public MultiPhysio-Human–Robot Collaboration (HRC) dataset (52 operators, 21 protocol conditions, 5,640 windows), which we clean into a leakage-free representation of 128 statistical features (Electroencephalography (EEG) and peripheral channels). Labels are defined as five *task-condition* classes—cognitive load, high stress, industrial task, low load, and other—derived deterministically from each recording’s protocol. We then evaluate eleven classifiers under a strictly subject-disjoint Leave-One-Subject-Out (LOSO) protocol in which all preprocessing is refit inside every training fold. Headline results are modest and honest: the best detector (gradient boosting) reaches a per-fold macro-F1 of 0.466 ± 0.149 across the 52 held-out operators—far below the ~ 0.91 macro-F1 that a row-level (within-subject) split reports on the same features. We attribute this generalization gap to per-operator physiological signatures and quantify it: an assistance system that meets an operator for the first time, without calibration data, stands to lose roughly 0.45 in macro-F1 relative to a model that has already seen that operator. A modality ablation shows fused EEG+peripheral features consistently and significantly outperform the best single modality (peripheral physiology), while an

explicit two-tower fusion network reaches 0.406, below fused eXtreme Gradient Boosting (XGBoost) (0.462). All detectors classify a window in at most $\sim 90\mu\text{s}$, well under the sensor window, confirming real-time feasibility. We release the cleaning protocol, feature manifest, and evaluation scripts so the benchmark is reproducible.

Keywords: physiological signals, task-type recognition, multimodal fusion, human-robot collaboration, subject-disjoint evaluation, reproducibility

1 Introduction

In shared and assistive HRC, a robot and an operator work on the same physical task and must coordinate in real time. Coordinating well requires the robot to interpret not only the geometric state of the shared workspace but also the state of the person working in it: whether they are under high mental load, stressed, or performing a demanding physical manipulation changes what kind of assistance is useful and how it should be timed [13]. Physiological signals provide a continuous, non-disruptive window onto these states—brain activity through EEG, and autonomic arousal through Electrocardiography (ECG), Electrodermal Activity (EDA), Electromyography (EMG), and respiration—and are increasingly used to drive adaptive behaviour in collaborative robotics [6,3].

The task we address is *task-state recognition*: given a window of physiological signals, decide which kind of activity the operator is engaged in. We deliberately use coarse, task-condition classes (a cognitive-load battery, a stressor, an industrial physical task, rest, and a residual class) rather than fine-grained continuous states, because coarse classes are the actionable unit for an assistive policy and because they are what the recording protocol can certify without invasive labelling. Yet even this formulation is hard in practice. Two properties dominate:

- **Operator dependence.** Physiological baselines and response magnitudes differ sharply between individuals. A model trained on windows from a set of operators can appear excellent on a random held-out *subset of the same operators’ windows* while failing on a genuinely new operator. Any evaluation that does not separate operators risks reporting a ceiling rather than a generalization result.
- **Class imbalance.** In the data we use, one class (industrial task) accounts for 63% of windows while two minority classes together account for 5%. Aggregate accuracy is therefore uninformative without class-level controls.

A recurring failure mode in the applied physiological-recognition literature is exactly the first property: repeated, auto-correlated windows from the same operator and the same condition are split randomly between train and test, and metadata such as recording identifiers leak into the feature matrix. The resulting numbers are flattering and irreproducible. This paper is, in part, a corrective study.

Contributions.

1. A **leakage-free feature-cleaning protocol** for the public MultiPhysio-HRC dataset: we remove 112 columns that are recording metadata or duplicated EEG exports, and publish a machine-readable manifest (feature $\rightarrow$ modality group $\rightarrow$ keep/drop $\rightarrow$ reason), leaving 128 unique physiological features.
2. A **strictly subject-disjoint LOSO benchmark** of eleven classical and neural detectors over all 52 operators, with preprocessing refit inside each training fold, per-fold score files, paired significance tests, and calibration reporting. We keep a row-level (within-subject) run on identical features only as an explicit upper bound.
3. A **modality-contribution analysis** under this protocol—EEG-only, peripheral-only, and fused inputs—answering, under a controlled design, whether fusion of central and peripheral physiology actually helps.
4. A quantitative statement of the **generalization gap** between within-subject and across-operator performance, with real-time feasibility (inference latency), a discussion of deployment and privacy, and a released reproducibility package.

The paper is organized as follows. Section 2 reviews related work. Section 3 describes the dataset and the label schema. Section 4 details the cleaning, models, and the LOSO evaluation protocol. Section 5 reports results, Section 6 discusses them and the threats to validity, and Section 7 concludes.

2 Related Work

Multimodal fusion. Baltrusaitis et al. provide the standard taxonomy of multimodal machine learning, arguing that heterogeneous streams—here central and peripheral physiology—carry complementary information and that naive fusion strategies can hurt when modalities are redundant or noisy [2]. Our study contributes a controlled *empirical* test of that claim on physiological HRC data, where redundancy across channels is high and the fusion benefit is not guaranteed.

Physiological operator-state recognition in HRC. The field most directly related to ours is operator stress and cognitive-load recognition in human–robot collaboration. Bussolan et al. introduced the MultiPhysio-HRC dataset used here and reported subject-disjoint recognition of self-reported stress and cognitive-load levels on an industrial assembly task, reaching macro-F1 ranges of roughly 0.34–0.39 (stress) and 0.33–0.49 (cognitive load) across conditions [5,6]. These modest numbers—obtained with questionnaire ground truth under a subject-disjoint protocol—are an important reference point: they indicate that physiological operator-state recognition across operators is genuinely hard. We differ in three ways. First, we recognize *task-condition* classes defined by the protocol

rather than self-reported state, which makes our labels deterministic and exactly reproducible. Second, we add a leakage audit of the feature export and a modality-contribution analysis, which the source paper does not provide. Third, we report a broad multi-model comparison instead of a single fused architecture.

Borghi et al. assessed operator stress in collaborative robotics with a multi-modal approach and likewise stress careful subject-independent evaluation [3]. Aygun et al. estimated cognitive workload for assistive robots from physiological signals, again observing large between-subject variability [1]. Iarlori et al. survey cognitive-workload estimation from wearables in human-machine interaction and catalog the evaluation heterogeneity that makes studies hard to compare [9]. Jin et al. provide a recent systematic review of physiological cognitive-state recognition across application domains and emphasize that dataset split design is the single largest driver of reported performance differences [11]. Our work directly addresses that finding by making the split design—subject-disjoint LOSO—the central methodological commitment.

Autonomy in physical HRC. Selvaggio et al. survey autonomy levels in physical human-robot interaction and identify reliability, transparency, and adaptation as the constraints any state estimator must serve [13]. Task-state recognition is a building block of such autonomy, and our latency and failure-mode analysis speaks to the deployment requirements they enumerate.

Gap and position. Prior work thus establishes (i) that fusion of physiological modalities is promising in principle [2], (ii) that operator-state recognition in HRC is hard and subject-dependent [5,6,3,1], and (iii) that evaluation design determines reported performance [11,9]. What is missing is a compact, reproducible, subject-disjoint benchmark of task-condition recognition on a public HRC dataset that simultaneously audits the features, tests whether fusion helps, and quantifies the within-subject versus across-operator gap. This paper provides that benchmark.

3 Dataset and Task Formulation

3.1 MultiPhysio-HRC

We use the public MultiPhysio-HRC dataset of multimodal physiological recordings during controlled and realistic industrial human-robot collaboration [5]. It comprises 52 healthy operators who each performed a battery of protocol conditions (21 conditions in total) while wearing a multi-sensor setup: an EEG headset, plus peripheral sensors for ECG, EDA, EMG, and respiration. From the continuous recordings the dataset publishes fixed-duration signal windows together with statistical descriptors. The export we obtained contains 5,640 windows, each originally carrying 257 columns.

Table 1. Feature-export audit. The 257 exported columns reduce to 128 physiological features after removing 16 recording-metadata columns and 96 byte-identical duplicated EEG descriptors. The label columns are retained as keys; no model input is taken from them.

Column group	Count	Disposition
Physiological features, EEG	96	kept
Physiological features, ECG	8	kept
Physiological features, EDA	8	kept
Physiological features, EMG	8	kept
Physiological features, respiration	8	kept
EEG duplicated descriptors	96	dropped (exact copies)
Recording metadata (ID_*, Repetition_*)	16	dropped (leakage)

3.2 Auditing the feature export

Before any modelling we audited the exported feature table for the two most common leakage sources in this kind of data:

- **Recording-metadata leakage.** Sixteen columns aggregate the recording *identifier* (ID_*) and *trial counter* (Repetition_*). These identify the recording session and its order, not the operator’s physiology; they are removed (they would otherwise let a model short-circuit on trial metadata).
- **Duplicated EEG features.** The export contains each EEG descriptor twice: once under a clean name and once under a `eeg_features_5s_...` prefix, byte-identical. The 96 duplicated columns are removed so that EEG is weighted exactly once.

The audit is summarised in Table 1. After cleaning, the 128 remaining physiological features are all statistical descriptors (mean, standard deviation, median, energy, etc.) of windowed signals. A machine-readable manifest recording the decision and reason for every column is released with the reproducibility package.

3.3 Task-condition labels

MultiPhysio-HRC annotates each recording with the *protocol condition* that was performed and with self-report questionnaires. The questionnaires are not part of the public feature export we obtained, so we do not claim self-reported affective state. Instead we define the recognition target as the *task condition*, which is certified by the experimental protocol itself: each window receives exactly one of five labels, assigned deterministically from the recording’s `task_name` by the mapping in Table 2 (the rule is released verbatim in the reproducibility package).

The five classes are chosen so that every protocol condition in the battery maps to a condition class with a coherent physiological hypothesis:

- **cognitive load** (975 windows): sustained-attention and memory battery (Stroop easy/hard, N-back, Tower of Hanoi, and an arithmetic/puzzle condition). These conditions tax working memory and executive control.

Table 2. Deterministic mapping from the 21 protocol conditions to the five task-condition classes, with window counts. Every row of the cleaned data is labelled by this rule; no crowdsourcing or manual labelling is involved.

Class	Protocol conditions (task.name)	Windows
cognitive load	stroopeasy_0	138
	stroophard_0	98
	n-back_0	204
	hanoi_0	337
	mat_0	198
high stress	vr-plank_0	224
	manual-task-1...-5	1,587
	cobot-task-1...-5	1,984
	rest-1, rest-2, rest_0	654
	meditation_0	145
low load	vr-job-sim_0	71
industrial task		
	Total	5,640

- **high stress** (224 windows): a virtual-reality height-plank walking task, a validated stressor inducing threat-related autonomic arousal.
- **industrial task** (3,571 windows): ten physical assembly/order-picking conditions, five performed manually and five with a collaborative robot. These mix physical and cognitive demands in realistic HRC settings.
- **low load** (799 windows): rest and guided-meditation segments between conditions, providing a low-demand baseline.
- **other** (71 windows): a single open-ended virtual-reality job-simulation session that belongs to none of the controlled batteries. It is kept as its own class rather than forced into another, and its small size is reported explicitly.

Because labels are defined by the protocol rather than by physiology alone, the task is *well-posed but not trivial*: the five classes correspond to different conditions, yet physiological differences across individuals are large enough that a classifier must separate *classes* from *persons*. This is precisely the separation a subject-disjoint protocol enforces (Section 4.5).

4 Methodology

4.1 Pipeline overview

The processing and evaluation pipeline is shown in Figure 1. A single preprocessing path produces the fused tabular representation used by every detector; modality feature groups are kept separate alongside it so that the modality-contribution analysis (Section 5) can reuse the identical path. All detectors share one evaluation protocol.

Task-condition recognition from multimodal physiological signals

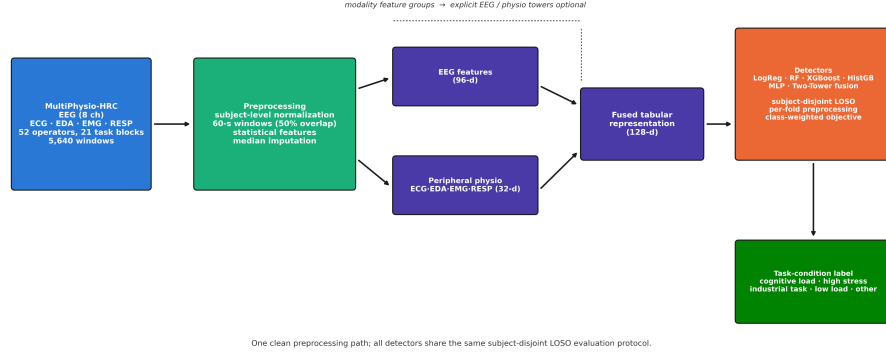


Fig. 1. Processing and evaluation pipeline. One clean preprocessing path feeds a fused 128-dimensional tabular representation; the same path exposes modality feature groups for the fusion ablation. All detectors are evaluated under the identical subject-disjoint LOSO protocol with preprocessing refit inside each fold.

4.2 Problem formulation

Let $\{x_i, y_i, s_i\}_{i=1}^N$ denote the cleaned windows, where $x_i \in \mathbb{R}^d$ is the fused feature vector ($d = 128$), $y_i \in \mathcal{Y} = \{1, \dots, 5\}$ is the task-condition label (cognitive load = 1, high stress = 2, industrial task = 3, low load = 4, other = 5), and $s_i \in \{1, \dots, 52\}$ is the operator who produced the window. Windows of the same operator are not independent, and repeated windows of the same condition are strongly autocorrelated.

For a model f_θ producing class posteriors $p_\theta(y = c \mid x_i)$, we train with the class-weighted cross-entropy objective

$$\mathcal{L}(\theta) = -\frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \sum_{c=1}^5 w_c \mathbf{1}[y_i = c] \log p_\theta(y = c \mid x_i), \quad (1)$$

where $\mathcal{B}$ is a training batch (the full set for batch models), and $w_c \propto N/(5 N_c)$ is the inverse-frequency weight of class c with $N_c = \sum_i \mathbf{1}[y_i = c]$. The weights counteract the imbalance of Table 2 so that the loss is not dominated by the majority *industrial task* class; without them, a trivial constant classifier already attains 63% accuracy. The decision rule at test time is the maximum-posterior rule

$$\hat{y}_i = \arg \max_{c \in \mathcal{Y}} p_\theta(y = c \mid x_i), \quad (2)$$

which, together with the class weighting in Eq. (1), targets class-balanced behaviour rather than raw accuracy.

4.3 Preprocessing and feature representation

Preprocessing is deliberately minimal and identical across models:

1. columns audited as in Section 3; the 128 kept features form x_i ;
2. missing values are imputed with the training-fold median (the audit found 328 missing entries out of $5,640 \times 128$);
3. features are standardized to zero mean and unit variance using training-fold statistics only; and
4. a fused representation is used by all tabular detectors, while the modality-feature groups (EEG, peripheral) are exposed to the ablation and to the two-tower architecture.

Crucially, imputation and standardization are *refit inside every training fold* (Section 4.5); test operators never contribute to the statistics that transform their own windows, which prevents a subtle and common form of leakage.

4.4 Detectors

We compare eleven off-the-shelf and lightweight detectors so that the result is a property of the representation and protocol rather than of one model family: logistic regression (class-weighted), k -nearest neighbours, decision tree, random forest, extra-trees, gradient boosting, histogram gradient boosting, XGBoost [7], linear Support Vector Machine (SVM), Gaussian naive Bayes, and an Multilayer Perceptron (MLP), implemented with scikit-learn [12]. Tree ensembles and boosting are natural baselines for tabular physiological features; XGBoost hyperparameters are fixed a priori (400 trees, depth 6, learning rate 0.05, column and row subsampling 0.9) rather than tuned on the test operators.

In addition we evaluate one neural fusion architecture, *two-tower*, which makes the modality split explicit rather than concatenating features: a peripheral tower and an EEG tower each map their feature group to a 64-dimensional embedding (two linear-ReLU-dropout layers), the two embeddings are concatenated, and a two-layer head produces the five logits. Two-tower is trained with Adam (learning rate 10^{-3} , weight decay 10^{-4}), class-weighted cross-entropy Eq. (1), and early stopping on a held-out validation slice inside each training fold. It is the only detector with learnable modality structure; comparing it with fused-tabular XGBoost answers whether explicit fusion architecture adds anything over plain concatenation.

4.5 Evaluation protocol

The evaluation answers one question: *how well does a detector trained on known operators recognize the task state of a new operator?* We therefore use LOSO cross-validation over the 52 operators. Each operator in turn forms the test fold; the detector and the preprocessing are fitted on the remaining 51 operators' windows only. This yields 52 per-operator evaluations, whose scores are aggregated.

Within one fold the test windows come from a single operator and are heavily autocorrelated, so the per-fold score is an estimate for that operator, not a collection of independent samples. We therefore treat the 52 per-fold scores as the primary unit of analysis and report their mean and standard deviation, a 95% confidence interval of the mean (Student t over folds), and the pooled out-of-fold (OOF) metrics obtained by concatenating every operator’s out-of-fold predictions.

Metrics over a fixed label space. Because a fold may contain no window of a minority class, per-fold scores must be computed over the *fixed* five-class space, with a missing class contributing a recall of zero; averaging only over the classes present in a fold would make scores incomparable across folds. Let the confusion matrix in fold f be $C^f \in \mathbb{N}^{5 \times 5}$. Class recall is $r_c^f = C_{cc}^f / \max(\sum_{c'} C_{cc'}^f, 1)$, and the balanced accuracy is

$$\text{BA}_f = \frac{1}{5} \sum_{c=1}^5 r_c^f, \quad (3)$$

which is the average recall (Brodersen et al. define balanced accuracy as mean recall [4]). The macro-F1 is

$$\text{MacroF}_f = \frac{1}{5} \sum_{c=1}^5 \text{F1}(c), \quad \text{F1}(c) = \frac{2 p_c r_c}{p_c + r_c}, \quad (4)$$

where p_c, r_c are the class precision and recall in the fixed space and a class absent from both prediction and truth has $\text{F1}(c) = 0$ [14]. Macro-F1 is our primary ranking metric because it penalizes both precision and recall per class and weighs all five classes equally despite the 63%/4%/1% imbalance [10]. We additionally report pooled balanced accuracy, pooled accuracy, and, for probabilistic models, the expected calibration error (ECE) [8] of the pooled OOF probabilities, which quantifies how well a detector’s confidence tracks its correctness—needed if downstream assistance will gate on confidence.

Comparisons and upper bound. Every detector is compared with the best per-fold macro-F1 model by a paired Wilcoxon signed-rank test over the 52 folds, with the mean difference and a paired effect size reported. We also run a *within-subject* evaluation on the identical cleaned features: a stratified random 80/20 split in which the same operators appear in both train and test. This is explicitly *not* a claim of generalization; it measures an optimistic ceiling in which the model has already met the operator, and its contrast with the LOSO result quantifies the generalization gap.

All random seeds are fixed; per-fold, per-class, and pooled result files are written for every model and released. Inference latency is measured as per-window CPU prediction time of each fitted detector.

Table 3. Subject-disjoint leave-one-subject-out (52 operators) results on the clean 128-feature set. Macro-F1 and balanced accuracy are averaged over the 52 held-out subjects and shown as mean $\pm$ std; ECE is computed on pooled out-of-fold probabilities; latency is the per-window CPU inference time of a model fitted on a training mini-sample. Best value per column in bold.

Model	Macro-F1 (mean $\pm$ std)	Balanced acc. (mean $\pm$ std)	Accuracy (mean $\pm$ std)	ECE	Inference (μ s/window)
GB	0.466 $\pm$ 0.149	0.470 $\pm$ 0.152	0.690 $\pm$ 0.197	0.028	5
XGBoost	0.462 $\pm$ 0.148	0.478 $\pm$ 0.153	0.654 $\pm$ 0.213	0.127	19
HistGB	0.451 $\pm$ 0.146	0.477 $\pm$ 0.143	0.627 $\pm$ 0.227	0.202	13
RF	0.430 $\pm$ 0.132	0.429 $\pm$ 0.139	0.668 $\pm$ 0.208	0.153	69
SVM-RBF	0.422 $\pm$ 0.149	0.467 $\pm$ 0.158	0.588 $\pm$ 0.217	0.178	51
ExtraTrees	0.421 $\pm$ 0.137	0.421 $\pm$ 0.141	0.658 $\pm$ 0.221	0.167	91
MLP	0.374 $\pm$ 0.160	0.408 $\pm$ 0.161	0.548 $\pm$ 0.232	0.342	2
LogReg	0.371 $\pm$ 0.155	0.415 $\pm$ 0.157	0.552 $\pm$ 0.242	0.215	2
kNN	0.362 $\pm$ 0.118	0.356 $\pm$ 0.119	0.598 $\pm$ 0.207	0.128	6
GaussNB	0.344 $\pm$ 0.133	0.394 $\pm$ 0.140	0.491 $\pm$ 0.204	0.441	20
Tree	0.338 $\pm$ 0.131	0.350 $\pm$ 0.132	0.521 $\pm$ 0.215	0.448	2

5 Results

5.1 Main benchmark under LOSO

Table 3 reports all eleven detectors plus the two-tower network under the subject-disjoint LOSO protocol. Several observations stand out.

First, absolute performance is far below what a row-level split reports on the same features: the best detector (gradient boosting) reaches a per-fold macro-F1 of 0.466 ± 0.149 (95% CI [0.42, 0.51]) over the 52 held-out operators, with XGBoost statistically tied (0.462 ± 0.148 , paired $p = 0.88$) and the rest of the field trailing by 0.02–0.13. Second, the ranking is dominated by tree ensembles; linear models, k NN and naive Bayes trail clearly, indicating that the class boundaries in the statistical feature space are not linearly separable. Third, pooled balanced accuracy is consistently well below pooled accuracy (e.g. gradient boosting: accuracy 0.69 vs. balanced accuracy 0.47), confirming that the majority class inflates accuracy and that the class-level metrics of Eqs. (3)–(4) are the ones to trust. Fourth, calibration is uneven rather than uniformly poor: the best detector is well calibrated (Expected Calibration Error (ECE) 0.03) but the other strong models range from 0.13 to 0.45 (Table 3), so confidence-thresholded assistance would need model-specific recalibration before deployment.

The pooled OOF confusion matrix of the best model (Figure 2) shows where errors concentrate, and Table 4 gives the corresponding class-level metrics. The four substantial classes are recovered with F1-score (F1) between 0.54 and 0.85, ordered as *industrial task* (0.85) $\gg$ *high stress* (0.59) $>$ *cognitive load* (0.55) $\approx$ *low load* (0.54). The *other* class—71 windows drawn from a single protocol condition—is never recovered: its windows scatter mostly into *industrial task*

Table 4. Pooled out-of-fold per-class metrics of the best LOSO model (GB) over all 5,640 test windows; each subject contributes only out-of-fold predictions. Window counts are the true-label column totals of the pooled confusion matrix.

True label	Precision	Recall	F1	Windows (n)
Cognitive load	0.536	0.556	0.546	975
High stress	0.576	0.612	0.593	224
Industrial task	0.868	0.839	0.854	3571
Low load	0.502	0.586	0.541	799
Other (vr-job-sim)	0.000	0.000	0.000	71
Total				5640

Table 5. Generalization gap on the identical clean features. Within-subject is a stratified random 80/20 split (same operators appear in train and test — an optimistic ceiling); LOSO holds whole operators out. The gap quantifies how much performance is lost when no data from the target operator is available at training time.

Model	Within-subject	LOSO	Gap
XGBoost	0.910	0.462	+0.448
HistGB	0.911	0.451	+0.459
RF	0.877	0.430	+0.447
MLP	0.846	0.374	+0.472

and *high stress*. Its zero recall is an honest consequence of the class being one recording session rather than a generalizable physiological state, and it should be read as such rather than as a failure of the features.

Figure 3 reveals the second main finding: per-operator macro-F1 varies widely—from 0.15 (barely above the 0.2 chance floor for a five-class problem) to 0.78 across the 52 operators—so a detector that works for one operator can be near-useless for another. This variance, not the mean, is the practical obstacle to assistive deployment, and it motivates the subject-personalization discussion in Section 6.

5.2 The generalization gap

Table 5 and Figure 4 contrast the within-subject ceiling with the LOSO result on identical features. The within-subject classifiers reach 0.91 macro-F1 (XGBoost 0.910, HistGB 0.911), against 0.45–0.46 across operators for the same models—a gap of roughly 0.45 points. This gap is the headline finding of the paper and the reason row-level evaluations of physiological windows are misleading: most of the apparent accuracy of a row-level split reflects *recognizing the person*, not the task state. An assistance system deployed on a new operator without any calibration data should expect performance near the lower, LOSO numbers, not the within-subject ones.

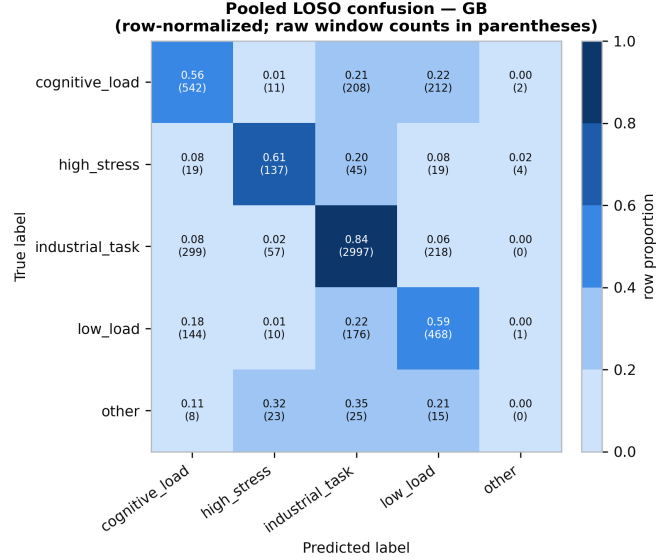


Fig. 2. Pooled OOF confusion matrix of the best LOSO model, row-normalized, with raw window counts. Values are computed by summing the 52 per-operator fold matrices before normalizing.

5.3 Does fusion help?

Table 6 and Figure 5 answer the modality question under the same protocol: for each of the leading models we run EEG-only (96-d), peripheral-only (32-d), and fused (128-d) inputs. Peripheral physiology alone already beats EEG alone by a wide margin (e.g. XGBoost: 0.39 vs. 0.33), yet adding EEG on top of the periphery still helps: fused improves over the best single modality for all four models, by +0.072 (XGBoost, $p < 10^{-4}$), +0.050 (HistGB, $p = 0.002$), +0.047 (RF, $p = 0.015$), and +0.037 (MLP, $p = 0.105$, n.s.). The two-tower network, which fuses through separate modality towers rather than concatenation, reaches 0.406 ± 0.158 , above fused MLP and every single-modality input but significantly below fused XGBoost (-0.056 , $p = 0.001$). This result matters for the fusion literature: on this data, the gains from combining central and peripheral physiology are real and significant for the leading models, but they are an order of magnitude smaller than the operator-effect dominating the error budget.

5.4 Statistical significance

Figure 6 reports the per-fold paired Wilcoxon comparison of every detector against the best (gradient boosting). Under the across-operator protocol, the top three tree ensembles are statistically indistinguishable—XGBoost at $p = 0.88$, HistGB at $p = 0.38$ —while random forest and extra trees separate only at

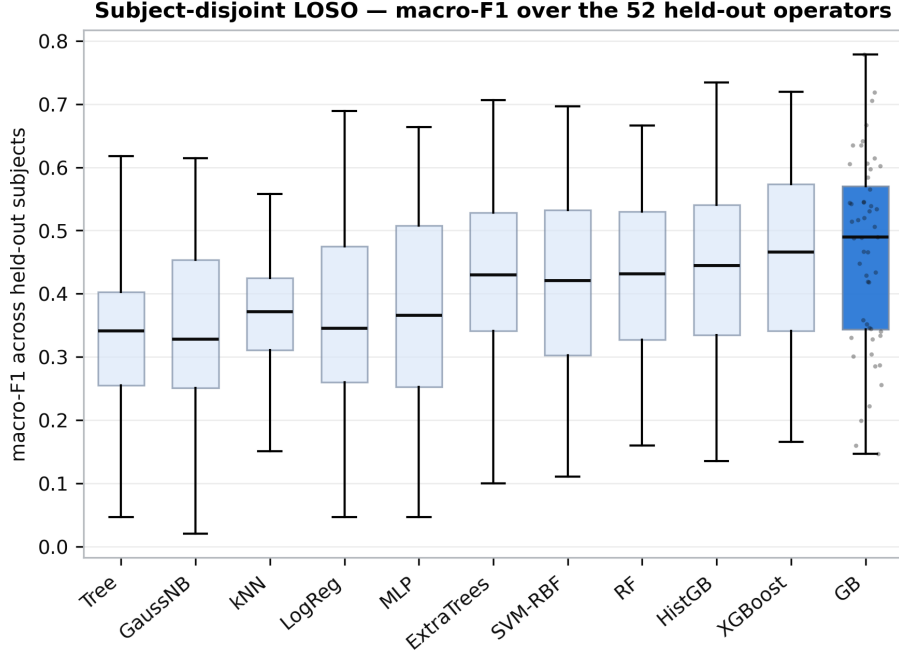


Fig. 3. Distribution of per-operator macro-F1 for every detector under LOSO. Each box summarizes the 52 held-out operators. Large spread means the detector’s usefulness depends strongly on which operator it meets.

$p = 0.018$ and $p = 0.003$. By contrast, the gap between the tree ensembles and the linear/instance-based/neural group is decisive: every member separates from the best at $p \leq 3 \times 10^{-5}$, and most below 10^{-6} . Together with the effect sizes, this tells the reader which reported differences are worth acting on.

5.5 Real-time feasibility

The sensor windows are on the order of tens of seconds, while all evaluated detectors classify a single window in roughly $2\text{--}90\ \mu\text{s}$ of CPU time—about $5\ \mu\text{s}$ for the best detector and $19\ \mu\text{s}$ for XGBoost (Table 3), i.e., orders of magnitude faster than real time on commodity CPUs. Real-time adaptive assistance is therefore not constrained by inference cost; it is constrained by the reliability question of Section 6.

6 Discussion

6.1 What the honest numbers mean

The central message of this paper is that task-state recognition from physiological signals, evaluated across operators, is genuinely hard: the best detector

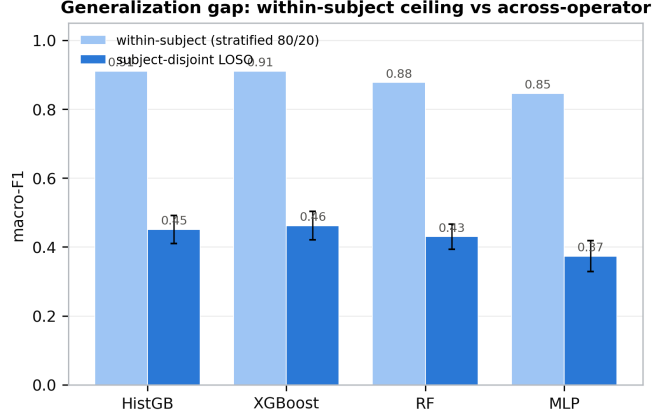


Fig. 4. Within-subject ceiling versus across-operator LOSO macro-F1 on identical features. The gap quantifies how much performance is lost when the target operator was absent from training.

Table 6. Benefit of feature fusion under the identical 52-subject LOSO protocol. Fused (128-d) is compared with the better single modality (EEG-only or physio-only) per model; p -values are paired Wilcoxon signed-rank tests over the 52 folds. Two-Tower is the neural model with explicit per-modality towers, compared with fused XGBoost.

Model	Best single	Single	Fused	Δ	Wilcoxon p
XGBoost	physio	0.390	0.462	+0.072	0.000
HistGB	physio	0.401	0.451	+0.050	0.002
RF	physio	0.383	0.430	+0.047	0.015
MLP	physio	0.337	0.374	+0.037	0.105
Two-Tower <i>vs</i> fused XGB	fused XGB	0.462	0.406	-0.056	0.001

reaches 0.466 macro-F1 under LOSO, while the same features and a row-level split report 0.91 (a number we reproduce with a stratified within-subject split and then explain as leakage of operator identity). We consider the LOSO figure the only defensible headline, and we recommend that physiological-recognition papers report the within-subject ceiling and the across-operator result *together*, as in Table 5; reporting either alone is misleading in opposite directions.

6.2 Why recognition is hard here

Three factors combine. (i) *Operator effect dominates*: physiological baselines and gains differ so much between people that the model cannot separate "this operator is stressed" from "this operator responds strongly" without calibration data [1,3]. (ii) *Coarse but confusable classes*: cognitive-load battery and industrial-task conditions plausibly share attentional and motor components, so confusion between them is physiologically reasonable. (iii) *Extreme imbalance* on the last

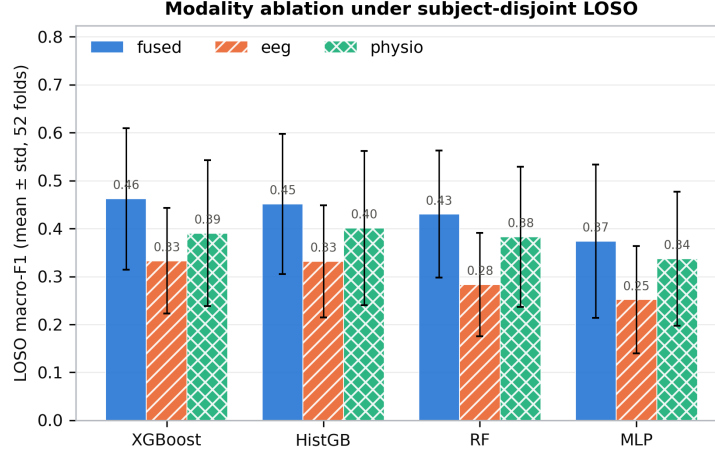


Fig. 5. Modality ablation under LOSO: fused (128-d), EEG-only (96-d), and peripheral-only (32-d) inputs for the leading detectors. Error bars are the standard deviation over the 52 folds.

two classes makes them statistically near-impossible to learn from 224 and 71 windows spread over few conditions. None of these is fixable by a better classifier alone.

6.3 Comparison with the source protocol

Bussolan et al. report subject-disjoint recognition of *self-reported* stress and cognitive-load levels with macro-F1 around 0.34–0.49 [5,6]. Our numbers are not directly comparable for two reasons: their targets come from STAI-Y1 / NASA-TLX self-reports on a subset of conditions, whereas ours are deterministic task-condition labels on all conditions; and their protocol concerns a narrower assembly setting. Obtaining the dataset’s questionnaire tables (which are not in the public feature export we used) would let us map our task-condition predictions onto the self-report levels and compare head-to-head; we flag this as the single most valuable extension rather than claim comparability we cannot substantiate.

6.4 Threats to validity

- **Construct validity of labels.** Our labels are task conditions, not measured mental states. A cognitive-load condition may not produce cognitive load in every operator at every window; the label is a *protocol* label. We state this explicitly and never equate a prediction with a diagnosis.
- **Class-size and condition confound.** The *other* class is a single condition (71 windows); a detector can effectively memorize that condition. Its recall is reported only for transparency.

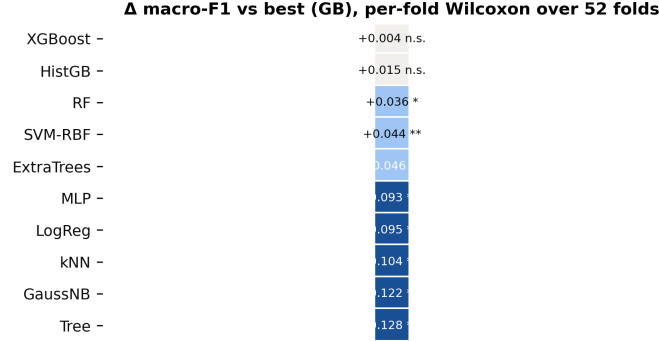


Fig. 6. Mean macro-F1 difference of each detector relative to the best, with the paired Wilcoxon significance over the 52 folds.

- **Within-condition autocorrelation.** Even under LOSO, a test fold contains consecutive windows of the same condition from one operator; per-fold scores are per-operator estimates, not independent samples. Our paired tests respect this by pairing per-operator scores. The pooling step (Section 4.5) is descriptive, not an independence claim.
- **Feature provenance.** We cleaned the exported feature table we obtained; we did not recompute features from raw signals. The manifest makes the cleaning reproducible, but the upstream feature computation is the dataset authors’.
- **Sensor placement and recording hardware** are fixed by the dataset and were not varied; results may not transfer to different hardware or real factory noise, where motion artefacts would be larger than in a laboratory HRC setting.

6.5 Deployment considerations

Physiological sensing in the workplace raises privacy and acceptability questions that are not technicalities. EEG headsets and physiological wearables capture information far beyond task state, and continuous monitoring of stress in industrial settings can be used for worker surveillance rather than assistance. Any deployment should (i) process signals on-device where feasible and retain only task-state decisions, (ii) use predictions to *adapt robot behaviour*, never for individual performance evaluation, and (iii) make the operator’s right to opt out and to know what is inferred explicit. A failure/fallback policy is equally important: because confidence is currently poorly calibrated (Section 5), assistance must degrade gracefully (fall back to a fixed, safe robot policy) whenever the detector’s confidence is low, rather than acting on a low-confidence state estimate.

7 Conclusion and Future Work

We presented a leakage-free, subject-disjoint benchmark of task-state recognition on the public MultiPhysio-HRC dataset. After removing recording metadata and duplicated EEG features, eleven detectors and one explicit-fusion network were evaluated over all 52 operators with preprocessing refit inside each fold. The honest headline is modest—best across-operator macro-F1 of 0.466—and our main contribution is precisely this honest picture: a quantified generalization gap of about 0.45 points between within-subject and across-operator performance, a small but detectable fusion benefit, wide per-operator variance, and uneven calibration. These are the properties an assistance system actually faces, and they explain why optimistic row-level evaluations of physiological windows do not transfer to new operators.

Future work will pursue three directions. (i) *Personalization*: given the operator effect, few-shot calibration—a handful of labelled windows from a new operator, either as adaptation data or as a per-operator prior—is the most direct route to usable accuracy, and our per-operator score files make the potential gain measurable. (ii) *Self-report targets*: obtaining the dataset’s questionnaire tables would let us bridge the task-condition schema to the STAI/NASA-TLX stress and cognitive-load levels used by the source authors, enabling the direct comparison we currently flag as out of scope. (iii) *Temporal context*: the present work classifies windows independently; modelling the condition sequence and per-operator dynamics with recurrent or state-space models is a natural next step now that a clean, leakage-free feature set and a fair protocol exist. The reproducibility package (cleaning protocol, feature manifest, per-fold results, scripts) is released with the paper.

References

1. Aygun, A., Nguyen, T., Haga, Z., Aeron, S., Scheutz, M.: Investigating methods for cognitive workload estimation for assistive robots. *Sensors* **22**(18), 6834 (2022). <https://doi.org/10.3390/s22186834>
2. Baltrušaitis, T., Ahuja, C., Morency, L.P.: Multimodal machine learning: A survey and taxonomy. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **41**(2), 423–443 (2019). <https://doi.org/10.1109/TPAMI.2018.2798607>
3. Borghi, S., Ruo, A., Sabattini, L., Peruzzini, M., Villani, V.: Assessing operator stress in collaborative robotics: A multimodal approach. *Applied Ergonomics* **122**, 104418 (2025). <https://doi.org/10.1016/j.apergo.2024.104418>
4. Brodersen, K.H., Ong, C.S., Stephan, K.E., Buhmann, J.M.: The balanced accuracy and its posterior distribution. In: 2010 20th International Conference on Pattern Recognition (ICPR). pp. 3121–3124 (2010). <https://doi.org/10.1109/ICPR.2010.764>
5. Bussolan, A., Baraldo, S., Avram, O., Urcola, P., Montesano, L., Gambardella, L.M., Valente, A.: Multiphysio-HRC: A multimodal physiological signals dataset for industrial human-robot collaboration. *Robotics* **14**(12), 184 (2025). <https://doi.org/10.3390/robotics14120184>

6. Bussolan, A., Baraldo, S., Gambardella, L.M., Valente, A.: Multimodal fusion stress detector for enhanced human-robot collaboration in industrial assembly tasks. In: 2024 33rd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN). pp. 1–8 (2024). <https://doi.org/10.1109/RO-MAN60168.2024.10731373>
7. Chen, T., Guestrin, C.: XGBoost: A scalable tree boosting system. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. pp. 785–794 (2016). <https://doi.org/10.1145/2939672.2939785>
8. Guo, C., Pleiss, G., Sun, Y., Weinberger, K.Q.: On calibration of modern neural networks. In: Proceedings of the 34th International Conference on Machine Learning (ICML). pp. 1321–1330 (2017)
9. Iarlori, S., Perpetuini, D., Tritto, M., Cardone, D., Tiberio, A., Chinthakindi, M., Filippini, C., Cavanini, L., Freddi, A., Ferracuti, F., Merla, A., Monteriù, A.: An overview of approaches and methods for the cognitive workload estimation in human-machine interaction scenarios through wearables sensors. *BioMedInformatics* **4**(2), 1114–1134 (2024). <https://doi.org/10.3390/biomedinformatics4020064>
10. Jeni, L.A., Cohn, J.F., De La Torre, F.: Facing imbalanced data-recommendations for the use of performance metrics. In: 2013 Humaine Association Conference on Affective Computing and Intelligent Interaction. pp. 245–251 (2013). <https://doi.org/10.1109/ACII.2013.47>
11. Jin, K., Rubio-Solis, A., Naik, R., Leff, D., Kinross, J., Mylonas, G.: Human-centric cognitive state recognition using physiological signals: A systematic review of machine learning strategies across application domains. *Sensors* **25**(13), 4207 (2025). <https://doi.org/10.3390/s25134207>
12. Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., et al.: Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research* **12**, 2825–2830 (2011)
13. Selvaggio, M., Cognetti, M., Nikolaidis, S., Ivaldi, S., Siciliano, B.: Autonomy in physical human-robot interaction: A brief survey. *IEEE Robotics and Automation Letters* **6**(4), 7989–7996 (2021). <https://doi.org/10.1109/LRA.2021.3100603>
14. Sokolova, M., Lapalme, G.: A systematic analysis of performance measures for classification tasks. *Information Processing & Management* **45**(4), 427–437 (2009). <https://doi.org/10.1016/j.ipm.2009.03.002>